Tianrui (Dennis) Wu

Curriculum Vitae

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Education

Jan. 2026–May 2027 — **University of Arizona** — *Tucson, Arizona, USA*
Bachelor of Science in Physics and Mathematics; expected May 2027.
GPA: 3.87/4.00.
Advisor: Prof. Tim Eifler, Arizona Cosmology Lab.
Dean's List; Global Wildcat Scholarship ($10,000/year merit award).

Aug. 2021–Dec. 2024 — **Duke University** — *Durham, North Carolina, USA*
Coursework toward a Bachelor of Science double major in Physics and Mathematics with a minor in Computer Science; transferred.
GPA: 3.92/4.00; Physics GPA: 4.00/4.00.
Advisors: Prof. Michael A. Troxel and Prof. Daniel Scolnic.
Dean's List with Distinction.

Publications

[1] **Wu, T.**, Paillas, E., et al. "BAO Reconstruction with Photometric Redshift Smearing." Manuscript in preparation (2026).

Peer-Reviewed Publications

[2] OpenUniverse, LSST Dark Energy Science Collaboration, Roman Project Infrastructure Teams, Alarcón, A., et al. (including **T. Wu**). "OpenUniverse2024: A Shared, Simulated View of the Sky for the Next Generation of Cosmological Surveys." *Monthly Notices of the Royal Astronomical Society* **544**(4), 3799–3823 (December 2025). doi:10.1093/mnras/staf1833.

[3] Riess, A. G., Scolnic, D., et al. (including **T. Wu**). "JWST Validates HST Distance Measurements: Selection of Supernova Subsample Explains Differences in JWST Estimates of Local H_0." *The Astrophysical Journal* **977**, 120 (2024). arXiv:2408.11770.

Research Experience

Jan. 2026–Present — **BAO Reconstruction with Photometric Redshift Smearing** — *University of Arizona*
Research Assistant, Arizona Cosmology Lab; mentor: Prof. Tim Eifler

- Lead a first-author study of baryon acoustic oscillation (BAO) reconstruction under photometric-redshift smearing as the sole analyst, coordinating with the principal investigator and a postdoctoral collaborator across the full pipeline from simulated-catalog generation through Bayesian posterior inference.
- Build a forward-model framework that propagates photometric-redshift uncertainty through reconstruction to quantify its effect on standard-ruler calibration for next-generation surveys including Roman and Rubin.
- Diagnosed and corrected a systematic bias in the Bayesian MCMC pipeline by cross-validating two covariance-estimation methods across a 250-cell sensitivity grid (5 noise levels $\times$ 5 data subsets $\times$ 2 model configurations), reducing the bias by sixfold.
- Mapped, using a 25-realization Monte Carlo ensemble, the smearing amplitude at which reconstruction's 20–40% precision gain becomes statistically insignificant; first-author manuscript in preparation.

Jun. 2024–Aug. 2024 — **Predicting Power-Spectrum Window Functions with Neural Networks** — *Caltech*
Summer Undergraduate Research Fellow, SPHEREx Team; mentor: Dr. Olivier Doré

- Built a ResUNet convolutional neural network in PyTorch to deconvolve instrument window-function effects from galaxy power spectra, reaching sub-percentage prediction error on validation data.
- Owned the full machine-learning pipeline: lognormal mock-catalog generation with realistic SPHEREx survey masks, distributed training on Linux HPC, and validation against analytic baselines.
- Improved the analysis pathway for local primordial non-Gaussianity $f_{\rm NL}^{\rm loc}$ and delivered a project proposal, two interim reports, a final presentation, and a final report to the SPHEREx faculty committee.

May 2022–Dec. 2024 — **Weak-Lensing Cosmology with Survey Synergies** — *Duke University*

Research Assistant, Cosmology Group; mentor: Prof. Michael A. Troxel

- Engineered a Python pipeline for joint statistical analysis of 120 TB of Roman Space Telescope and Rubin Observatory simulations, constructing two-point correlation functions and using MCMC-based Bayesian inference to constrain cosmological parameters.
- Evaluated mode-based strategies and ground–space survey synergies for photometric-redshift calibration; the new methods increased the Figure of Merit by 20% and tightened constraints on cosmic homogeneity.
- Contributed simulation and analysis work to the OpenUniverse2024 collaboration and its MNRAS publication.

Sep. 2023–Aug. 2024 — **Distance-Ladder Calibration with Synthetic Stellar Populations** — *Duke University*

Research Assistant, Cosmology Group; mentor: Prof. Daniel Scolnic

- Developed a Python pipeline for synthetic stellar catalogs and systematic feature extraction from Hertzsprung–Russell diagrams. Integrated novel distance-calibration methods into a unified framework for cross-team comparisons.
- Compared JWST distance measurements across the literature and quantified galaxy–dust correlations with current SDSS data using SQL and Python; contributed to the ApJ publication listed above.

May 2023–Aug. 2023 — **Quantifying Wetland Carbon Emissions in the Southeastern United States** — *Duke University*

Data+ Team Lead; mentor: Prof. Wenhong Li

- Led a four-person team that applied climatology-based adaptive missing-value imputation and optimized a random-forest model by grid search; mentored teammates developing SVM and multilayer-perceptron models.
- Delivered a layered R Shiny analytics application and final research presentation.

Industry and Quantitative Research

Aug. 2025–Present — **Quantitative Researcher and Systematic Trader** — *Qrigin Capital, Remote*

Full-time Aug. 2025–May 2026; part-time May 2026–Present. Proprietary trading firm ($35M)

- **Alpha and directional strategy:** Built an LSTM forecasting engine using phase-space features with a full MLflow MLOps lifecycle; integrated 200+ multi-frequency OHLCV, order-book, and derivatives factors with decay analysis and cross-sectional ranking, lifting live-model win rates by 3.1%. Deployed an HFT taker strategy on Coinbase International at a post-fee Sharpe of 8.2 over three months.
- **Execution and market making:** Architected TWAP, VWAP, and implementation-shortfall algorithms with microstructure timing overlays, reducing realized slippage by approximately 2 bps versus TWAP. Built vectorized and L2 queue-position backtesting engines used to validate a grid market-making strategy at a three-month out-of-sample Sharpe of 9.
- **Trading operations and risk:** Ran CTA books across OKX, Binance, and Coinbase while owning P&L monitoring, transaction-cost analysis, and fee analysis. Directed multi-strategy risk across $35M in prop/client capital using Grafana monitoring, leverage, position, and beta limits plus pre/post volatility and model attribution; trimmed exposure ahead of the October market crash.
- **Research leadership:** Led a two-person research pod across production code, factor research, microstructure analysis, market-making models, and strategy development; mentored junior researchers and served as the primary methodology and code reviewer before live deployment.

Applied technologies: Python (PyTorch, MLflow, Polars, pandas, NumPy), CEX REST/WebSocket APIs, Linux, Git

Feb. 2025–Aug. 2025 — **Quantitative Research Intern** — *InsightCheck Investments, Shenzhen, China*

Systematic equity fund with CTA desk ($300M assets under management)

- Developed a market-neutral long/short equity strategy combining PCA-derived statistical factors, mean-reversion signals, and a covariance-based risk model, backtested at a post-cost Sharpe of 2.4.
- Researched fundamental and microstructure alpha factors for Chinese A-shares, validating 62+ intraday factors with daily information coefficient above 0.01 and conducting cross-sectional portfolio optimization in Polars.
- Solely engineered an L3 market-by-order reconstruction engine, prototyping in Python and migrating to C++, achieving approximately one second per trading day for the full A-share universe.
- Built a Dask-based distributed backtester that shortened evaluation cycles tenfold to approximately 15 minutes.

Applied technologies: Python (Polars, Dask), C++, SQL, Linux, Git

Selected Quantitative Projects

2026 — **IMC Prosperity 4 Trading Competition** — *Python (PyTorch), Rust*

- Ranked in the top 3% of 6,000+ teams worldwide by developing an automated trading bot using behavioral fingerprinting of counterparties, fair-value estimation, market making, and cross-asset arbitrage.

2025 — **Crypto Market Maker and Backtest Engine** — *Python, Rust*

- Built and deployed a live ETH market-making framework on OKX that combined EWMA flow alphas with a CNN capturing order-book patterns through phase-space features, supported by a WebSocket data feed and Grafana monitoring.
- Averaged 0.5–1% daily return; the project led to recruitment at Qrigin Capital.

Awards and Fellowships

2026 — **National First Place, U.S. Physicists' Tournament**

2026 — **Global Wildcat Scholarship, University of Arizona**
$10,000/year merit scholarship for international transfer students

2024 — **Faculty Scholars Finalist, Duke University**
Sole physics nominee; among Duke's highest undergraduate honors

2024 — **Jocelyn Bell Burnell Outstanding Leadership Scholarship, Society of Physics Students**
$6,000; sole recipient nationwide

2024 — **Inductee, Sigma Pi Sigma National Physics Honor Society**
Membership awarded for academic distinction in physics

2023 — **Outstanding Chapter Award, Society of Physics Students**
Awarded to the Duke chapter during Wu's presidency

2021 — **Third Grand Award, Physics and Astronomy, Regeneron International Science and Engineering Fair**
$1,000 award in the Physics and Astronomy category

2021 — **Top Gold Medalist, British Physics Olympiad**
Ranked in the national top ten

2021–2026 — **Dean's List with Distinction / Full-Time Dean's List**
Duke University (Fall 2021, Spring 2023, Fall 2023, Fall 2024) and University of Arizona (Spring 2026)

Talks and Presentations

Jun. 2026 — **BAO Measurements, Cosmology Session**
SPHEREx Science Team Meeting (Face-to-Face E4), IPAC, Caltech

Apr. 2026 — **BAO analysis segment of the SPHEREx team colloquium**
SO/NSF NOIRLab Joint Colloquium, Steward Observatory, University of Arizona

Aug. 2024 — **Predicting Power Spectrum Window Function with Neural Networks**
SURF Final Presentation, SPHEREx Team, Caltech

Teaching

Feb. 2025–Aug. 2025 — **Physics and Mathematics Tutor** — *PalmDrive Academy*
Taught high-school physics and mathematics for 20 hours/week, including AP Physics, AP Calculus, AP Mathematics, and Cambridge A-Level Physics; guided more than 15 students to the top AP exam score of 5.

Aug. 2022–Apr. 2024 — **Teaching Assistant and SAGE Facilitator** — *Duke University Mathematics Department*
Supported Multivariable Calculus, Linear Algebra, and Advanced Probability. Assisted 40+ students weekly across two three-hour Help Room shifts and led two one-hour SAGE discussion sections each week. Rated "Excellent" (30/30) by supervisors and peers in the 2023 and 2024 reviews.

Leadership and Outreach

Sep. 2021–Dec. 2024 | **Founder and President, Stargazing Devils Astronomy Club** | *Duke University*

- Founded Duke's first astronomy club and organized weekly observatory open houses with 100+ attendees, three comet events, four meteor-shower events, and three eclipse events; served as a NASA Partner Eclipse Ambassador.
- Secured funding for 14 students to attend the 2023 and 2024 solar eclipses, organized a campus eclipse watch party for 300+ participants, and led 20+ astrophotography events using the Skynet platform.

Jun. 2022–Dec. 2024 | **President (2024), Co-President (2023), Society of Physics Students** | *Duke University*

- Revived the chapter after the pandemic through four research panels, six student–faculty seminars, three laboratory tours, MATLAB/LaTeX workshops, graduate-school panels, and a women's mentorship program.
- Organized the 2023 SPS Zone 5 meeting for 100+ participants; the chapter received the 2023 Outstanding Chapter Award.

Jun. 2022–May 2024 | **Student Ambassador, American Physical Society** | *Remote*

- Recruited APS members; organized physics study sessions, manuscript workshops, and publication events; served as Technology Chair for the Physics Undergraduates Learning and Sharing Experiences webinar series.
- Attended the 2023 and 2024 APS Leadership Meetings and advocated on physics-community priorities during U.S. Congressional Visits Day.

Sep. 2021–Aug. 2024 | **Senior Editor and Peer Reviewer, Duke Vertices Science Journal** | *Duke University*

Reviewed two STEM manuscripts per month and implemented a synopsis system for general audiences that increased website traffic by 210%; invited by the editor-in-chief to join the executive team.

Technical Skills, Coursework, Languages, and Interests

Programming	Python (expert), Rust, C/C++, Java, R, SQL/PostgreSQL, MATLAB, Julia, JavaScript, and LaTeX
Machine Learning and Data	PyTorch, MLflow, Polars, pandas, NumPy, Numba, Dask, qlib, SciPy, astropy, Matplotlib; ResUNet/CNN and LSTM architectures; random forests, support-vector machines, multilayer perceptrons, feature engineering, selection, and grid-search optimization
Statistical and Scientific Methods	Bayesian inference, MCMC, Monte Carlo simulation, statistical modeling, sensitivity analysis, covariance estimation, two-point correlation functions, information-coefficient validation, distributed training and backtesting, Linux HPC, Git, REST/WebSocket APIs, and Grafana monitoring
Domain Knowledge	Observational cosmology (BAO, weak lensing, power spectra, photometric-redshift calibration), market microstructure, live strategy development and backtesting, optimal execution, factor research, digital assets, commodity futures, global equities, options (Akuna 101/201 completed), and prediction markets
Selected Coursework	Stochastic Calculus; Statistical Machine Learning; Advanced Mathematical Modeling; Computational Physics; Real Analysis; Advanced Probability; Statistical Learning and Inference; Graduate Machine Learning; Probability; Data Structures and Algorithms; Microeconomics; Macroeconomics; Game Theory; Cosmology*; General Relativity*; Stellar Astrophysics*; Quantum Theory; Electrodynamics; Abstract Algebra*; Differential Geometry *Graduate-level course.
Languages	English (fluent); Mandarin Chinese (native)
Interests	Personal trading portfolio (AI-assisted, factor-driven strategy with discretionary macro and catalyst overlays); poker; competitive GeoGuessr; tabletop wargaming; strategy board games; chaos theory; forecasting; astrophotography; cosmology; and science fiction